

Demo Script

Condensed, copy-paste-ready run sheet for the live CI walkthrough

A condensed, copy-paste-ready run sheet for the live walkthrough — same content as [docs/WALKTHROUGH.md](#), formatted for scanning quickly while presenting or following along. About 10 minutes end to end.

Before starting

- ☐ On `main`, terminal open, the repository's **Actions** tab open in a browser tab.
- ☐ Confirm the baseline:

COPY THIS BLOCK

```
git checkout main
git pull
./mvnw clean test
```

Expected: Tests run: 15, Failures: 0, Errors: 0.

1. Baseline (~1 min)

COPY THIS BLOCK

```
./mvnw clean test
```

Tests run: 15, Failures: 0 in well under a second — no Docker, no database, no network call. That's what makes the rest of this run fast.

2. Branch, then break the build on purpose (~2 min)

COPY THIS BLOCK

```
git checkout -b break-the-build
```

In `src/main/java/com/example/portfoliodemo/Portfolio.java`, inside `buy(...)`, comment out:

COPY THIS BLOCK

```
cash = cash.subtract(cost);
```

COPY THIS BLOCK

```
./mvnw clean test
```

Expected: Tests run: 15, Failures: 2 — PortfolioTest.buyDeductsCashBalance and PortfolioTest.sellIncreasesCashBalance. One bug, two broken behaviors: this is why a suite covers more than one scenario.

3. Push and open a pull request (~2 min)

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```
git add -A
git commit -m "Introduce a bug on purpose for the CI demo"
git push -u origin break-the-build
gh pr create --fill
```

(No `gh`? Use the “Compare & pull request” banner GitHub shows on the repository page after the push.)

4. Watch the check fail (~2 min)

- Refresh the PR page: the check goes yellow (queued/running) → red, in about 15–30 seconds (a JVM build takes a little longer to spin up than an interpreted one).
- **Details** on the failed check → same test output as local.
- **Merge** button: greyed out — “Merging is blocked — Required statuses must pass before merging.” Applies to everyone, including repository admins — that’s the branch protection rule on `main`.

5. Fix it and watch it go green (~2 min)

Uncomment the line:

COPY THIS BLOCK

```
cash = cash.subtract(cost);
```

COPY THIS BLOCK

```
./mvnw clean test    # confirm 15 tests pass locally first
git add -A
git commit -m "Fix the cash balance bug"
git push
```

Refresh the PR: check turns green → **Merge** button active → merge (`Squash and merge` is a fine default).

6. Recap (~1–2 min)

- This whole loop took minutes only because the suite is fast and infrastructure-free — a slow one trains people to stop waiting for it, which is how bugs like this reach `main` on real projects.
- Branch protection turns “please run the tests before merging” into “you cannot merge until they pass” — enforced, not requested.
- The entire pipeline is `.github/workflows/ci.yml` — versioned and reviewed like any other code.

If something stalls

- Actions runs typically take 20–40s to go from “queued” to a result for a JVM project (downloading dependencies, starting the JVM) — that gap is normal, not a failure.
- If a local test result looks stale right after an edit (still red after a fix, or still green after breaking something), run `./mvnw clean test` instead of `./mvnw test` — Maven can occasionally reuse a compiled class from just before the edit. This doesn’t happen on GitHub Actions, which always builds from a clean checkout.
- Branch protection can be turned off from **Settings** → **Branches** if `main` needs to be unblocked for any reason.

For every step explained in full, with diagrams, see [docs/WALKTHROUGH.md](#). To repeat this same exercise on your own fork afterwards, see its [“Try it yourself”](#) section.